```
lab_10=# CREATE TABLE departments (
lab_10(#     department_id SERIAL PRIMARY KEY,
lab_10(#     department_name VARCHAR(100) NOT NULL,
lab_10(#     building VARCHAR(100),
lab_10(#     budget NUMERIC(12, 2)
lab_10(# );
CREATE TABLE
lab_10=#
lab_10=#
lab_10=# CREATE TABLE students (
lab_10(#     student_id SERIAL PRIMARY KEY,
lab_10(#     first_name VARCHAR(50) NOT NULL,
lab_10(#     last_name VARCHAR(50) NOT NULL,
lab_10(#     email VARCHAR(100) UNIQUE NOT NULL,
lab_10(#     enrollment_year INTEGER,
lab_10(#     gpa NUMERIC(3, 2),
lab_10(#     department_id INTEGER REFERENCES departments(department_id)
lab_10(# );
CREATE TABLE
lab_10=#
lab_10=# CREATE TABLE courses (
lab_10(#     course_id SERIAL PRIMARY KEY,
lab_10(#     course_name VARCHAR(100) NOT NULL,
lab_10(#     credits INTEGER NOT NULL,
lab_10(#     department_id INTEGER REFERENCES departments(department_id)
lab_10(# );
CREATE TABLE
```

```
lab_10=# CREATE TABLE enrollments (
lab_10(#     enrollment_id SERIAL PRIMARY KEY,
lab_10(#     student_id INTEGER REFERENCES students(student_id),
lab_10(#     course_id INTEGER REFERENCES courses(course_id),
lab_10(#     semester VARCHAR(30),
lab_10(#     grade VARCHAR(2),
lab_10(#     UNIQUE (student_id, course_id, semester)
lab_10(# );
CREATE TABLE
lab_10=#
lab_10=# INSERT INTO departments
lab_10-# (department_name, building, budget)
lab_10-# VALUES
lab_10-# ('Software Engineering', 'Main Building', 150000.00),
lab_10-# ('Computer Science', 'Technology Center', 180000.00),
lab_10-# ('Business Administration', 'Business Building', 120000.00),
lab_10-# ('Economics', 'Main Building', 95000.00),
lab_10-# ('Mathematics', 'Science Building', 110000.00),
lab_10-# ('Media and Communications', 'Arts Building', 85000.00);
INSERT 0 6
lab_10=# INSERT INTO students
lab_10-# (first_name, last_name, email, enrollment_year, gpa, department_id)
lab_10-# VALUES
lab_10-# ('Amina', 'Sadykova', 'amina.sadykova@example.com', 2024, 3.85, 1),
lab_10-# ('Timur', 'Asanov', 'timur.asanov@example.com', 2023, 3.45, 2),
lab_10-# ('Dana', 'Kim', 'dana.kim@example.com', 2024, 3.92, 1),
lab_10-# ('Arman', 'Bekov', 'arman.bekov@example.com', 2022, 3.20, 3),
lab_10-# ('Aigerim', 'Nur', 'aigerim.nur@example.com', 2023, 3.75, 4),
lab_10-# ('Dias', 'Orozov', 'dias.orozov@example.com', 2024, 3.60, 2),
lab_10-# ('Madina', 'Aliyeva', 'madina.aliyeva@example.com', 2022, 3.95, 1),
lab_10-# ('Nursultan', 'Erkinov', 'nursultan.erkinov@example.com', 2023, 3.10, 5),
lab_10-# ('Alina', 'Petrova', 'alina.petrova@example.com', 2024, 3.80, 6),
lab_10-# ('Ruslan', 'Ismailov', 'ruslan.ismailov@example.com', 2022, 3.55, 2),
lab_10-# ('Sabina', 'Tulegenova', 'sabina.tulegenova@example.com', 2023, 3.88, 1),
lab_10-# ('Adilet', 'Mambetov', 'adilet.mambetov@example.com', 2024, 3.40, 3),
```

```
lab_10=# INSERT INTO courses
lab_10-# (course_name, credits, department_id)
lab_10-# VALUES
lab_10-# ('Introduction to Programming', 4, 1),
lab_10-# ('Database Systems', 4, 1),
lab_10-# ('Web Development', 3, 1),
lab_10-# ('Data Structures and Algorithms', 4, 2),
lab_10-# ('Artificial Intelligence', 4, 2),
lab_10-# ('Business Management', 3, 3),
lab_10-# ('Microeconomics', 3, 4),
lab_10-# ('Linear Algebra', 4, 5),
lab_10-# ('Statistics', 3, 5),
lab_10-# ('Digital Media', 3, 6);
INSERT 0 10
lab_10=#
lab_10=# INSERT INTO enrollments
lab_10-# (student_id, course_id, semester, grade)
lab_10-# VALUES
lab_10-# (1, 1, 'Fall 2025', 'A'),
lab_10-# (1, 2, 'Fall 2025', 'A'),
lab_10-# (1, 3, 'Fall 2025', 'B+'),
lab_10-# (2, 4, 'Fall 2025', 'B'),
lab_10-# (2, 5, 'Fall 2025', 'B+'),
lab_10-# (3, 1, 'Fall 2025', 'A'),
lab_10-# (3, 2, 'Fall 2025', 'A'),
lab_10-# (3, 5, 'Fall 2025', 'A'),
lab_10-# (4, 6, 'Fall 2025', 'B'),
lab_10-# (4, 7, 'Fall 2025', 'B+'),
lab_10-# (5, 7, 'Fall 2025', 'A-'),
lab_10-# (5, 9, 'Fall 2025', 'B+'),
lab_10-# (6, 4, 'Fall 2025', 'A-'),
lab_10-# (6, 5, 'Fall 2025', 'B+'),
lab_10-# (7, 1, 'Fall 2025', 'A'),
lab_10-# (7, 2, 'Fall 2025', 'A'),
lab_10-# (7, 3, 'Fall 2025', 'A'),
```

```
lab_10=# SELECT
lab_10-#     table_schema,
lab_10-#     table_name
lab_10-# FROM information_schema.tables
lab_10-# WHERE table_schema = 'public'
lab_10-#   AND table_type = 'BASE TABLE'
lab_10-# ORDER BY table_name;
 table_schema | table_name
--------------+-------------
 public       | courses
 public       | departments
 public       | enrollments
 public       | students
(4 rows)
```

```
lab_10=# SELECT
lab_10-#     ordinal_position,
lab_10-#     column_name,
lab_10-#     data_type,
lab_10-#     character_maximum_length,
lab_10-#     is_nullable
lab_10-# FROM information_schema.columns
lab_10-# WHERE table_schema = 'public'
lab_10-#   AND table_name = 'students'
lab_10-# ORDER BY ordinal_position;
 ordinal_position |   column_name   |     data_type     | character_maximum_length | is_nullable
------------------+-----------------+-------------------+--------------------------+-------------
                1 | student_id      | integer           |                          | NO
                2 | first_name      | character varying |                       50 | NO
                3 | last_name       | character varying |                       50 | NO
                4 | email           | character varying |                      100 | NO
                5 | enrollment_year | integer           |                          | YES
                6 | gpa             | numeric           |                          | YES
                7 | department_id   | integer           |                          | YES
(7 rows)
```

```
lab_10=# SELECT
lab_10-#     tc.table_name,
lab_10-#     tc.constraint_name,
lab_10-#     tc.constraint_type,
lab_10-#     kcu.column_name,
lab_10-#     ccu.table_name AS referenced_table,
lab_10-#     ccu.column_name AS referenced_column
lab_10-# FROM information_schema.table_constraints AS tc
lab_10-# LEFT JOIN information_schema.key_column_usage AS kcu
lab_10-#     ON tc.constraint_name = kcu.constraint_name
lab_10-#     AND tc.table_schema = kcu.constraint_schema
lab_10-# LEFT JOIN information_schema.constraint_column_usage AS ccu
lab_10-#     ON tc.constraint_name = ccu.constraint_name
lab_10-#     AND tc.table_schema = ccu.constraint_schema
lab_10-# WHERE tc.table_schema = 'public'
lab_10-#   AND tc.table_name IN (
lab_10(#       'departments', 'students', 'courses', 'enrollments'
lab_10(#   )
lab_10-#   AND tc.constraint_type IN ('PRIMARY KEY', 'FOREIGN KEY')
lab_10-# ORDER BY tc.table_name, tc.constraint_type;
 table_name  |       constraint_name       | constraint_type |  column_name  | referenced_table | referenced_column
-------------+-----------------------------+-----------------+---------------+------------------+-------------------
 courses     | courses_department_id_fkey  | FOREIGN KEY     | department_id | departments      | department_id
 courses     | courses_pkey                | PRIMARY KEY     | course_id     | courses          | course_id
 departments | departments_pkey            | PRIMARY KEY     | department_id | departments      | department_id
 enrollments | enrollments_student_id_fkey | FOREIGN KEY     | student_id    | students         | student_id
 enrollments | enrollments_course_id_fkey  | FOREIGN KEY     | course_id     | courses          | course_id
 enrollments | enrollments_pkey            | PRIMARY KEY     | enrollment_id | enrollments      | enrollment_id
 students    | students_department_id_fkey | FOREIGN KEY     | department_id | departments      | department_id
 students    | students_pkey               | PRIMARY KEY     | student_id    | students         | student_id
(8 rows)
```

```
lab_10=# SELECT
lab_10-#     s.student_id,
lab_10-#     s.first_name,
lab_10-#     s.last_name,
lab_10-#     s.email,
lab_10-#     s.gpa,
lab_10-#     d.department_name
lab_10-# FROM students AS s
lab_10-# JOIN departments AS d
lab_10-#     ON s.department_id = d.department_id
lab_10-# ORDER BY s.student_id;
 student_id | first_name | last_name  |             email              | gpa  |     department_name
------------+------------+------------+--------------------------------+------+--------------------------
          1 | Amina      | Sadykova   | amina.sadykova@example.com     | 3.85 | Software Engineering
          2 | Timur      | Asanov     | timur.asanov@example.com       | 3.45 | Computer Science
          3 | Dana       | Kim        | dana.kim@example.com           | 3.92 | Software Engineering
          4 | Arman      | Bekov      | arman.bekov@example.com        | 3.20 | Business Administration
          5 | Aigerim    | Nur        | aigerim.nur@example.com        | 3.75 | Economics
          6 | Dias       | Orozov     | dias.orozov@example.com        | 3.60 | Computer Science
          7 | Madina     | Aliyeva    | madina.aliyeva@example.com     | 3.95 | Software Engineering
          8 | Nursultan  | Erkinov    | nursultan.erkinov@example.com  | 3.10 | Mathematics
          9 | Alina      | Petrova    | alina.petrova@example.com      | 3.80 | Media and Communications
         10 | Ruslan     | Ismailov   | ruslan.ismailov@example.com    | 3.55 | Computer Science
         11 | Sabina     | Tulegenova | sabina.tulegenova@example.com  | 3.88 | Software Engineering
         12 | Adilet     | Mambetov   | adilet.mambetov@example.com    | 3.40 | Business Administration
         13 | Karina     | Volkova    | karina.volkova@example.com     | 3.70 | Economics
         14 | Eldar      | Osmonov    | eldar.osmonov@example.com      | 3.25 | Mathematics
         15 | Zarina     | Akhmetova  | zarina.akhmetova@example.com   | 3.97 | Software Engineering
         16 | Bekzat     | Toktogulov | bekzat.toktogulov@example.com  | 3.65 | Computer Science
         17 | Mira       | Abdullaeva | mira.abdullaeva@example.com    | 3.50 | Media and Communications
         18 | Samir      | Kadyrov    | samir.kadyrov@example.com      | 3.78 | Business Administration
         19 | Elina      | Semenova   | elina.semenova@example.com     | 3.90 | Economics
         20 | Ilyas      | Duisheev   | ilyas.duisheev@example.com     | 3.30 | Mathematics
(20 rows)
```

```
lab_10=# SELECT
lab_10-#     s.first_name,
lab_10-#     s.last_name,
lab_10-#     c.course_name,
lab_10-#     e.semester,
lab_10-#     e.grade
lab_10-# FROM enrollments AS e
lab_10-# JOIN students AS s
lab_10-#     ON e.student_id = s.student_id
lab_10-# JOIN courses AS c
lab_10-#     ON e.course_id = c.course_id
lab_10-# ORDER BY s.last_name, c.course_name;
 first_name | last_name  |          course_name           | semester  | grade
------------+------------+--------------------------------+-----------+-------
 Zarina     | Akhmetova  | Artificial Intelligence        | Fall 2025 | A
 Zarina     | Akhmetova  | Introduction to Programming    | Fall 2025 | A
 Madina     | Aliyeva    | Database Systems               | Fall 2025 | A
 Madina     | Aliyeva    | Introduction to Programming    | Fall 2025 | A
 Madina     | Aliyeva    | Web Development                | Fall 2025 | A
 Timur      | Asanov     | Artificial Intelligence        | Fall 2025 | B+
 Timur      | Asanov     | Data Structures and Algorithms | Fall 2025 | B
 Arman      | Bekov      | Business Management            | Fall 2025 | B
 Arman      | Bekov      | Microeconomics                 | Fall 2025 | B+
 Nursultan  | Erkinov    | Linear Algebra                 | Fall 2025 | B
 Nursultan  | Erkinov    | Statistics                     | Fall 2025 | B+
 Ruslan     | Ismailov   | Artificial Intelligence        | Fall 2025 | A-
 Ruslan     | Ismailov   | Data Structures and Algorithms | Fall 2025 | B+
 Dana       | Kim        | Artificial Intelligence        | Fall 2025 | A
 Dana       | Kim        | Database Systems               | Fall 2025 | A
 Dana       | Kim        | Introduction to Programming    | Fall 2025 | A
 Adilet     | Mambetov   | Business Management            | Fall 2025 | B+
 Aigerim    | Nur        | Microeconomics                 | Fall 2025 | A-
 Aigerim    | Nur        | Statistics                     | Fall 2025 | B+
 Dias       | Orozov     | Artificial Intelligence        | Fall 2025 | B+
 Dias       | Orozov     | Data Structures and Algorithms | Fall 2025 | A-
```

```
lab_10=# SELECT 'departments' AS table_name, COUNT(*) AS total_records
lab_10-# FROM departments
lab_10-# UNION ALL
lab_10-# SELECT 'students', COUNT(*) FROM students
lab_10-# UNION ALL
lab_10-# SELECT 'courses', COUNT(*) FROM courses
lab_10-# UNION ALL
lab_10-# SELECT 'enrollments', COUNT(*) FROM enrollments;
 table_name  | total_records
-------------+---------------
 departments |             6
 students    |            20
 courses     |            10
 enrollments |            30
(4 rows)
```

```
lab_10=# SELECT
lab_10-#     d.department_name,
lab_10-#     COUNT(s.student_id) AS number_of_students,
lab_10-#     ROUND(AVG(s.gpa), 2) AS average_gpa
lab_10-# FROM departments AS d
lab_10-# LEFT JOIN students AS s
lab_10-#     ON d.department_id = s.department_id
lab_10-# GROUP BY d.department_id, d.department_name
lab_10-# ORDER BY number_of_students DESC;
     department_name      | number_of_students | average_gpa
--------------------------+--------------------+-------------
 Software Engineering     |                  5 |        3.91
 Computer Science         |                  4 |        3.56
 Business Administration  |                  3 |        3.46
 Mathematics              |                  3 |        3.22
 Economics                |                  3 |        3.78
 Media and Communications |                  2 |        3.65
(6 rows)
```